Probabilistic Control Barrier Functions

Safety in Probability for Discrete-Time Stochastic Systems

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-Paper Summary by Prajith Saravanan

I. Introduction

Limitations of Existing Work

- Most prior literature assumes disturbances are **deterministic and bounded** with known limits.
- This assumption rarely holds in practice and leads to **overly conservative controllers**.

Motivation

- The authors design controllers that stay safe under **stochastic and potentially unbounded disturbances**.
- Existing discrete-time approaches often rely on **martingale theory** to bound the probability of leaving the safe set.
- These martingale-based methods typically use only the **first moment (mean)**, which can be too conservative , or simply inapplicable if the mean is unknown.

II. Contributions of the Paper

Contribution 1 - Introduces a **probabilistic notion of CBFs**.

> *Caveat: verifying whether a function is a probabilistic CBF is impractical in general, since it requires knowing the full uncertainty distribution.*

Contribution 2 - Verification via Moments: Two sufficient conditions guarantee that a function is a probabilistic CBF using only moments of the distribution of the CBF's value at the next time step.

- **Condition 1:** uses an upper bound on the CBF together with the **first moment (mean)**.
- **Condition 2:** uses the **first and second moments (mean and variance)**.

The paper also identifies when each condition is convex, making them tractable for optimization-based control.

Contribution 3 - Data-Driven Verification: When moment information is unavailable, the authors use **concentration inequalities and scenario optimization** to verify probabilistic CBFs directly from a dataset of uncertainty realizations (samples), giving conditions that hold with high confidence.

Contribution 4 — Validation: The theoretical results are demonstrated in both **simulation and real-world hardware** experiments using quadruped robots.

Contribution 1 - Problem Setup and Probabilistic CBFs

Discrete-Time Stochastic System

$\mathbf{x}_{t+1} = \mathbf{F}(\mathbf{x}_t, \mathbf{u}_t, \mathbf{d}_t)$

- $\mathbf{d}_t$ is a random variable drawn from a distribution D.
- For simplicity, D is assumed to be state- and time-independent.

Safe Set and Objective

- **Safe set:** a user-prescribed set $C = \{x \in \mathbb{R}^n : h(x) \geq 0\}$.
- **Objective:** design a state-feedback controller k so that the closed-loop trajectory stays inside C.

Achieving absolute safety is impossible whenever D is unbounded , the trajectory will eventually leave C with probability 1. The authors therefore target safety over a finite time horizon with a prescribed probability.

Definition 1 - Probabilistic Safety over a Finite Horizon

$$\mathbb{P}\Big(\bigcap_{t=0}^{H}\{\mathbf{x}_t \in \mathcal{C}\}\Big) \geq 1 - \epsilon$$

The probability that all states remain inside the safe set is softened to $1 - \varepsilon$. When this holds, the controller k is called ε-safe over a time horizon H.

Definition 2 - Probabilistic CBF

For a standard discrete-time CBF, the condition is $h(x_{t+1}) \geq \alpha \cdot h(x_t)$. Here, since the next state depends on a random disturbance d, the exact next state cannot be guaranteed:

$$\mathbb{P}\Big(h(\mathbf{F}(\mathbf{x}, \mathbf{u}, \mathbf{d})) \geq \alpha h(\mathbf{x})\Big) \geq 1 - \delta$$

h is called a δ-probabilistic CBF

Another definition which would be commonly used, $\Delta h(\mathbf{x}, \mathbf{u}, \mathbf{d}) = h(\mathbf{F}(\mathbf{x}, \mathbf{u}, \mathbf{d})) - \alpha h(\mathbf{x})$

> *Interpretation: at every state, there must exist a control action u that makes it highly probable (probability at least $1 - \delta$) that the safety condition holds at the very next time step.*

- Definition 1 covers the entire multi-step trajectory.
- Definition 2 gives a rule for a single step.

The paper proves that stringing together enough of these highly-safe single steps (Definition 2) achieves the multi-step safety goal (Definition 1).

Proposition 1

$$\mathbb{P}\Big(\bigcap_{t=0}^{H}\{\mathbf{x}_t \in \mathcal{C}\}\Big) \geq (1 - \delta)^H$$

Guarantees that designing a controller from the probabilistic CBF condition (Def. 2) at every step secures safety of the entire finite-time trajectory:

From Definition 1, if $\delta \leq 1-(1-\epsilon)^{\frac{1}{H}}$ holds, the controller k is ε-safe over a horizon H.

- Proposition 1 shows that controllers satisfying Def. 2 automatically satisfy the probabilistic-safety notion of Def. 1.
- **Limitation:** verifying that h is a probabilistic CBF per Def. 2 is impractical in practice, since it requires computing a probability under D, which is usually unknown.

Contribution 2 - Verification via Known Moments

Proposition 2 - Markov-Based Condition (First Moment + Upper Bound)

$$\mathbb{E}[\Delta h(\mathbf{x},\mathbf{u},\mathbf{d})] \geq b(1-\delta)$$

If **b > 0 is a known upper bound** ie, $\Delta h(x,u,d) \leq b$ on the safety margin and the mean condition above holds, then h is provably a δ-probabilistic CBF.

In implementation, such a condition is naturally added as a constraint in an optimization problem for the controller - however, this condition is generally not convex.

Making the Markov Condition Computable (Convexity)
The condition becomes tractable if either:

- **1) h is convex** -via Jensen's inequality applied to Proposition 2, and the following holds true,

$$: \tilde{\Delta} h(x,u) := h(\mathbb{E}[F(x,u,d)]) - \alpha h(x) \geq b(1-\delta)$$

or

- **2) h is concave**, with the Hessian of h bounded above by λ, ie , $\sup_{\mathbf{x}\in\mathbb{R}^n} \left\|\nabla^2 h(\mathbf{x})\right\| \leq \lambda$ and:

$$\hat{\Delta} h(\mathbf{x},\mathbf{u}) := \tilde{\Delta} h(\mathbf{x},\mathbf{u}) - \frac{\lambda}{2} Tr(Cov(\mathbf{F}(\mathbf{x},\mathbf{u},\mathbf{d}))) \geq b(1-\delta)$$

Then h is a δ-probabilistic CBF.

> *Neither equation is universally convex - they become tractable only under specific structure. Examples:*
>
> *If h is quadratic, the convex condition forms a QCQP ie, Quadratically constrained Quadratic Program (efficient heuristics exist).*
>
> *If F is affine in u with additive disturbance (F(x,u,d) = $F_1(x) + F_2u + d$) and h is concave, the concave-case condition becomes strictly convex.*

Proposition 4 - Cantelli-Based Condition (First and Second Moments)

If $\mathbb{E}[\Delta h(\mathbf{x},\mathbf{u},\mathbf{d})] - \sqrt{\mathrm{Var}\left(\Delta h(\mathbf{x},\mathbf{u},\mathbf{d})\right)\frac{1-\delta}{\delta}} \geq 0$, then h is a δ-probabilistic CBF.

Making this computable again requires convexity:

- **1) If h is convex and** $\tilde{\Delta} h(\mathbf{x},\mathbf{u}) \geq \sqrt{\mathrm{Var}\left(\Delta h(\mathbf{x},\mathbf{u},\mathbf{d})\right)\frac{1-\delta}{\delta}}$

or

- **2) If h is concave**, with the Hessian of h bounded above by λ, ie , $\sup_{\mathbf{x}\in\mathbb{R}^n} \left\|\nabla^2 h(\mathbf{x})\right\| \leq \lambda$ and:

$$\hat{\Delta} h(\mathbf{x},\mathbf{u}) \geq \sqrt{\mathrm{Var}\left(\Delta h(\mathbf{x},\mathbf{u},\mathbf{d})\right)\frac{1-\delta}{\delta}}.$$

Then h is a δ-probabilistic CBF.

> *The variance term Var(Δh(x,u,d)) can further complicate convexity.*
>
> *But if d is additive and h is affine, Var(Δh(x,u,d)) = Var(h(d)), removing its dependence on u.*
>
> *Notably, when Δh is affine in u, the condition becomes a second-order cone (SOC) constraint - which is convex and usable directly in an SOCP(Second order cone program)*

Contribution 3 - Data-Driven Verification

The moment-based conditions above require knowing statistical moments (mean, variance) of the uncertainty. In practice, engineers often only have raw data - a dataset of N realizations of the disturbance, $D_N = \{d^{(1)}, \ldots, d^{(N)}\}$.

Definition 3 - Probabilistic CBF with a Given Confidence

δ-probabilistic CBF with confidence 1-β

$$\mathbb{P}_N\left(\mathbb{P}(\Delta h(\mathbf{x},\mathbf{u},\mathbf{d}) \geq 0) \geq 1-\delta\right) \geq 1-\beta.$$

$D_N=\{d_{(1)},\ldots,d_{(N)}\}$ and Where P_N is the probability wrt D_N

> *Reading: "Based on the dataset we collected, we are (1 − β) confident that our system has a (1 − δ) chance of staying safe at the next step."*

Proposition 6 - Safety over a Finite Horizon (Using Data)

If you choose $\delta \leq 1-(1-\epsilon)^{\frac{1}{H}}$ as the single step parameter and $\beta \leq \frac{\tilde{\beta}}{H}$, the controller is ε-safe over horizon H with confidence 1 − β̃, written as:

$$\mathbb{P}_N\left(\mathbb{P}\left(\bigcap_{t=0}^{H}\{\mathbf{x}_t \in \mathcal{C}\}\right) \geq 1-\epsilon\right) \geq 1-\tilde{\beta}.$$

> *Intuition: to guarantee a total multi-step confidence, the required single-step confidence must be stricter , scaled by the horizon length H.*

Proposition 7 - Hoeffding-Based Condition

Verifies the "double probability" condition using the empirical mean of the dataset, provided the safety margin Δh has a specific separable structure:

$$\Delta h(\mathbf{x},\mathbf{u},\mathbf{d}) = H_1(\mathbf{x},\mathbf{u}) + H_2(\mathbf{x},\mathbf{d}) + H_3(\mathbf{x},\mathbf{u})H_4(\mathbf{x},\mathbf{d})$$

All component functions H_i must be bounded: there exist constants a, b, a_2, b_2, a_4, b_4, b_3 satisfying:

$a \leq \Delta h(x, u, d) \leq b$,

$a_2 \leq H_2(x, d) \leq b_2$,

$a_4 \leq H_4(x, d) \leq b_4$, and $|H_3(x, u)| \leq b_3$ (almost surely)

For a chosen error margin ε, the sample size N must satisfy two exponential bounds from Hoeffding's inequality:

$$\frac{\beta}{2} \geq 2\exp\{-\frac{2N\epsilon_1^2}{(b_2-a_2)^2}\} \text{ and } \frac{\beta}{2} \geq 2\exp\{-\frac{2N\epsilon_2^2}{(b_4-a_4)^2}\}$$

If the above holds, and a control input u can be found such that the average safety margin across all N samples satisfies:

$$\frac{1}{N}\sum_{i=1}^{N}\Delta h(\mathbf{x},\mathbf{u},\mathbf{d}^{(i)}) \geq \epsilon + b(1-\delta)$$

Then h is a δ-probabilistic CBF with confidence 1 − β.

Proposition 8 - Scenario Approach-Based Condition

Relies on optimizing the controller against every data sample simultaneously.

If: $\mathbf{u} \rightarrow -\Delta h(\mathbf{x},\mathbf{u},\mathbf{d})$ *is convex*

and N satisfies $\sum_{i=0}^{d-1} \binom{N}{i} \delta^i (1-\delta)^{N-i} \leq \beta$ or equivalently(apparently) $N \geq \frac{2}{\delta}(\ln \frac{1}{\beta} + d)$, then h

is a δ-probabilistic CBF with confidence 1 - β.

> *Intuition: "If a control action perfectly survives all N recorded disturbance scenarios, what is the probability it survives a brand-new, unseen disturbance?"*
>
> *Scenario-approach theory dictates exactly how many samples N are needed to guarantee a failure rate no worse than δ with confidence 1 − β.*

Contribution 4 -Validation

This section validates the theoretical solvers developed in Contributions 2,3 by applying them to a quadrupedal robot tasked with navigating a narrow corridor

The Dynamics: The robot is modeled using discrete-time unicycle-like dynamics with a disturbance term: :

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} v_k^x \\ v_k^y \\ \omega_k \end{pmatrix} + \mathbf{d}_k,$$

where, $x_k=[x,y,\theta]^T$ and $\Delta t>0$

The Safety Set

The narrow corridor is defined by **h(x) = 0.5² − y²**, which strictly confines the robot's lateral y-position between **−0.5 m and 0.5 m**.

A. Simulation

- **Setup:** Stochasticity is simulated via a Gaussian disturbance in the y-direction, d□ ~ N(0, σ). Time horizon **H = 20** steps, single-step failure rate **δ = 0.1**, confidence **β = 0.01**, and **α = 0.01**.
- **Solvers Compared:** A nominal controller (no safety filter) is tested against five approaches **Markov, Cantelli, Hoeffding, Scenario**, and an existing **Martingale-based** method from prior literature.
- **Data & Optimization:** The data-driven Hoeffding and Scenario approaches use datasets of **3,252** and **113** samples, respectively. At each step, the safety filter solves an optimization problem:
  - **Markov, Hoeffding, and Cantelli** form convex **QCQPs**, solved in **~3–4 ms** using `cvxpy`.

    - The **Scenario approach**, despite also being convex, is markedly slower (**~80 ms**) due to its larger constraint set.
    - The **Cantelli** condition is non-convex and is instead solved with a **trust-region solver**, yet still runs efficiently (~3.8 ms).
- **Results - Conservatism:** Across 400 simulated trajectories, all probabilistic-CBF approaches produced **far fewer safety violations** than their theoretical bounds required. This conservatism stems from **limited knowledge of the true distribution**, use of **Jensen's inequality** in the proofs, and evaluating safety **step-by-step rather than over the full trajectory**.
- **Results - Trade-offs:** There is a clear **safety vs. feasibility trade-off**. Methods with the fewest violations (**Markov, Cantelli**) become **infeasible at smaller disturbance variances (σ)**. The older **Martingale** method tolerates larger variances but **fails to meet the prescribed safety guarantees** in practice.